

ORIGINAL RESEARCH

Learning global image representation with generalized-mean pooling and smoothed average precision for large-scale CBIR

Jinliang Yao^{1,2} | Yongqing Li¹  | Bing Yang¹ | Chenrui Wang¹

¹School of Computer Science and Technology,
Hangzhou Dianzi University, Hangzhou, China

²Zhejiang Key Laboratory of Brain-Machine
Collaborative Intelligence, Hangzhou, China

Correspondence

Yongqing Li, No. 1158, Ave. 2, Xiasa Higher
Education Park District, Hangzhou, Zhejiang
310018, China.
Email: liyongqing@hdu.edu.cn

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Abstract

Content-based image retrieval (CBIR) is the problem of searching for items in an image database that are similar to the query image. Most of the existing image retrieval methods are trained based on metric learning loss functions (e.g. contrastive loss or triplet loss), however, which require the use of hard sample mining strategies (HMS) to better train the model. The HMS implies that picking out hard positive or negative samples increases the complexity of model training and requires a large amount of additional training time. To address this issue, lessons from recent work are leveraged on representation learning and a model called GS is proposed that combines the state-of-the-art Generalized-Mean (GeM) pooling and the smoothed average precision (AP). The entire network can be learned end-to-end by approximating the non-differentiable AP function to a differentiable one-without mining hard samples, only image-level annotations. A model named GSA is also presented which achieves excellent retrieval performance jointly trained by two various loss functions. Experimental results validate the effectiveness of the proposed approach and demonstrate the competitive performance on a common standard image retrieval dataset (Revisited Oxford and Paris).

1 | INTRODUCTION

One of the most fundamental and significant vision tasks is content-based image retrieval, which is a long-standing field of research [1]. Image retrieval aims to find images that semantically match or are similar to the query image in a large image retrieval gallery analyzing the visual content of the image. For instance, if you have a picture of an item you want to buy on Amazon, you could upload the picture to Amazon, then the system would instantly return the relevant ones you want to buy. The above scenario illustrates how essential content-based image retrieval technology is in our real lives. Content-based image retrieval has also other applications that show great promise, such as person re-identification [2], remote sensing [3] and online market shopping recommendations [4]. A prevalent retrieval task is landmark retrieval [5, 6], which aims to pick out images containing a particular landmark from a database of millions of images. Since landmark retrieval datasets are heavily affected by illumination variations, viewpoint/pose, occlusion,

and clutter, the learned features need to be robust enough to tackle these complicated cases.

Representations (also known as descriptors) of images, which are employed to encode the content of an image, play a core role in a retrieval task. And the encoding is used to measure similarity between query images and the images in the retrieval database, so the ability to learn strong image descriptors is of utmost significance. Over the past two decades, significant progress has been made in image representation learning. Previous inchoate approaches relied mainly on hand-crafted local features, such as SIFT [7, 8]. Philbin et al. proposed a novel bag-of-words (BoW) model using SIFT features [9]. The problem of aggregating local features (such as SIFT) into global representations has also been widely studied, with the most famous ones being VLAD [10] and Fisher Vectors [11]. Owing to the development of deep learning techniques, traditional keypoint-based approaches have recently been outperformed by features extracted from deep models [12, 13]. The task of image retrieval is currently dominated by activation-valued image descriptors

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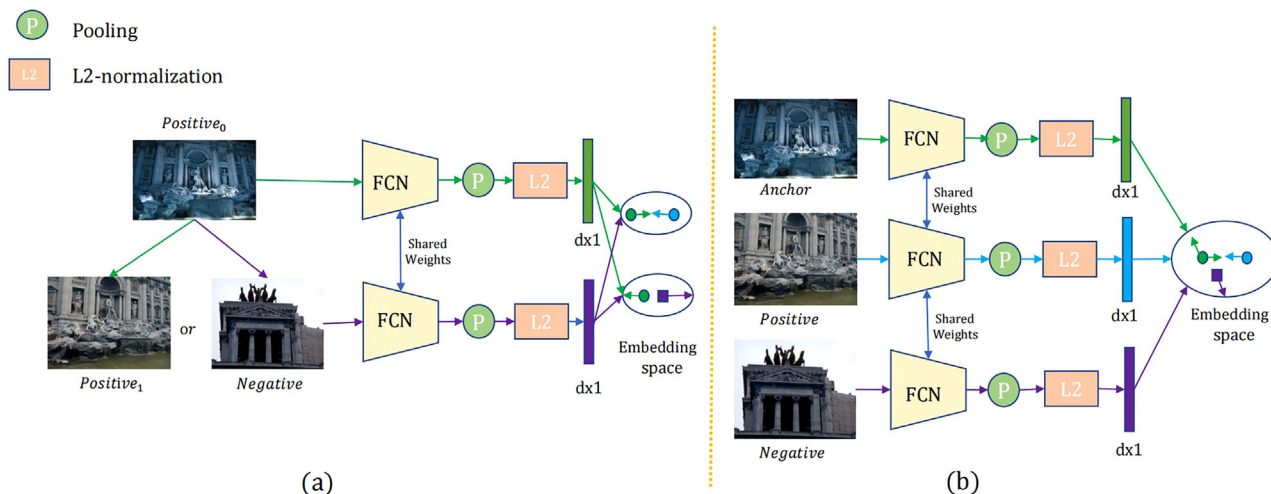


FIGURE 1 The typical architecture of metric learning of (a) two-branch siamese and (b) triplet network.

based on convolutional neural networks. The activation-valued image descriptors have shown superior discriminative ability and compactness [14]. Babenko et al. proposed a method to extract image descriptors by AlexNet [15] in which the last two fully-connected layers were used as feature extractors [14]. At first, the loss functions for classification tasks were applied to the retrieval tasks, which has become a performance bottleneck for image retrieval. Later, image retrieval was regarded as a problem of metric learning, that is, an image embedding was learned so that Euclidean distance could capture the similarity of image pairs better. The typical architecture of metric learning is two-branch Siamese [16, 17] (Figure 1a) or triplet network [18, 19] (Figure 1b). Instead of models aimed for classification, the above network structures can be exploited to accomplish the retrieval task by constructing the corresponding image pairs or triples. Recent models trained using contrastive loss [20–22] or triplet loss [23, 24] as the objective functions have made great progress on large-scale retrieval datasets. However, training high-performance and robust models using Siamese networks or triplet networks needs to mine the hard negative samples, which increases the complexity of model training. The hard negative samples refer to the mismatched images with the most similar image descriptor, and vice versa. To address the above issue, some researchers view the retrieval task from a different perspective, namely, image retrieval can not only be regarded as a metric learning problem but also as a ranking one. It seems more reasonable that the idea of ranking, which is closely related to the measurement metric of image retrieval (mean Average Precision), is employed to perform this visual task.

The essence of image retrieval can be formulated as a ranking problem [25], whose goal is to sort the images in an image retrieval gallery in descending order of similarity to the query image. The metrics of retrieval quality include AP, Recall, F-score, and Normalized Discounted Cumulative Gain (NDCG). However, AP has become the *de facto* standard metric for measuring retrieval quality [26]. Since the formula for direct optimization of AP contains a discrete ranking function that is neither differentiable nor decomposable, end-to-end train-

ing using gradient descent-based methods is impossible. Thus, directly optimizing average precision is notoriously difficult, the solution is to approximate the AP function into a smooth differentiable one.

Therefore, to circumvent the disadvantages of deep metric learning loss, this paper proposes an approach that combines the state-of-the-art Generalized-Mean (GeM) pooling [20] and smoothed average precision (Smooth-AP) [27] for learning global image representation. In addition, we also investigate the impact of using average query expansion (AQE) [12, 23, 28–30] and α -weighted query expansion (α QE) [20] on the accuracy of retrieval. Extensive ablation experiments are also carried out to explore the influence of each potential factor (e.g. the dimensionality of image features and descriptor whitening) on the final retrieval accuracy. Besides, inspired by [31, 32], we also combine two different kinds of loss functions for joint optimization. To summarize, our main contributions are as follows: (1) We propose a model called GS that can extract compact image representations for large-scale image retrieval. (2) We also introduce the model GSA, which is trained jointly using different loss functions. (3) Compared with previous studies, our method does not need to mine hard samples, which makes model training simpler and more efficient. Our models are able to perform a single-stage search and achieve impressive retrieval performance on ROxf and RPar datasets.

The rest of the paper is structured as follows: Section 2 discusses the related work, Section 3 describes the proposed methodology, Section 4 shows a series of experimental studies, Section 5 presents our discussion and limitations, and Section 6 presents our conclusions.

2 | RELATED WORK

In this section, we review related works from the following aspects: local features, global features, joint local and global features, direct optimization of AP, and attention mechanisms.

2.1 | Local feature

Prior to deep learning, SIFT [7] and SURF [33] are two of the well-known purely hand-crafted local features. They are first detected and then aggregated into a compact vector. In earlier work, the local descriptors were clustered by BoW using a pre-defined codebook $C = \{c_1, c_2, \dots, c_n\}$ with N centroids, which maps each local feature to the closest visual word. For each centroid c_n , one can count and normalize the number of occurrences. The BoW representation is a histogram of the visual words, so its dimension is equal to the number of cluster centres. Local features are also aggregated into a compact image representation by VLAD [10]. To improve the precision of visual word matching, Yao et al. proposed a novel contextual descriptor that measures the contextual similarity of visual words [34]. In addition, spatial validation using local features with RANSAC [35] and Aggregated Selective Matching Kernel (ASMK) [36] has shown significant performance gains. With the rise of deep learning, impressive progress has been made [21, 22, 37]. DELF [37] is a precursor and foundational work, which proposes an attention mechanism module to pick out key points and select local features with high confidence scores to measure the similarity between images. Tolias et al. proposed a model called HoW which integrated ASMK technology with DELF to enhance the model's retrieval ability [21]. Weinzaepfel et al. improved the HoW model and retrieval procedures to tackle the redundancy of numerous local features and the discrepancies between the training and testing stages [22].

2.2 | Global feature

Early approaches to applying CNNs for image retrieval set the fully-connected layer activations to be the global image descriptors [14]. A fully-connected layer has a global receptive field, which leads to two obvious limitations for image retrieval: a lack of spatial information and a lack of local geometric invariance. To address this issue, researchers move the focus to the activations of convolutional layers followed immediately by a global pooling operation. Therefore, a compact global descriptor with a dimensionality equivalent to the number of feature maps of the corresponding convolutional layer is yielded by employing a global pooling operation. These global-pooling methods include max pooling MAC [13], sum pooling SPoC [28] and the weighted sum pooling CroW [29]. Furthermore, R-MAC [12] can be regarded as a hybrid scheme. The features extracted from the local areas (in a manner of the sliding window) are performed by max pooling and sum pooling and ℓ_2 -normalization respectively to obtain the final compact image descriptor. We can infer from our existing knowledge that global max pooling preserves more texture information while global average pooling retains more background information. To consider the tradeoff between max pooling and average pooling, Generalized-Mean pooling (GeM) [20] has been proposed and has shown significant performance gains. Yang et al. proposed an end-to-end deep orthogonal local and global framework that can integrate local and global information inside images into

compact image representations [38]. The proposed model in [24] is jointly optimized using first-order and second-order loss functions, which is essentially based on a triplet loss.

2.3 | Joint local and global feature

It is natural to take into account both local and global features. Deep global features contain more abstract semantic and coarse-grained information, which is a high-level abstract representation of the whole image. While deep local features contain descriptors and geometric information about particular image regions. These local features are more frequently derived from visually sensitive regions. In general, an image has only a single global feature and multiple local features. Jointly learning local matching and global representations is beneficial to both sides. Tolias et al. trained a model with global features, and during the testing stage the top 1000 most prominent (high attention score) local features were selected for ASMK matching [21]. DELG [39] has been developed by Cao et al. that can be trained end-to-end with local and global features. During the testing stage, the global features are first applied for sorting, and then the top 100 images sorted in the first stage are re-ranked with local features.

2.4 | Optimizing average precision

The methods that directly optimize the AP metric by using listwise loss require to simultaneously consider a larger batch of images. The trend toward direct optimization of non-differentiable AP has recently seen a resurgence in the retrieval community. Sophisticated methods [40] have been developed to overcome the challenges of non-decomposability and non-differentiability in optimization. Many recent methods [25] have demonstrated that the differentiability of histogram bins can be utilized to approximate the AP. The loss function used in our proposed model adopts an approach with a sigmoid function substituting the non-differentiable indicator function. Although methods for optimizing AP have made great strides in information retrieval, they are greatly neglected by the visual community because of their complexity. In the realm of information and image retrieval, we probably foresee the development of simpler and more efficient loss functions that approximate AP. The community of vision will be interested in techniques that directly optimize AP.

2.5 | Attention mechanisms

Since the attention mechanism in [41] has been applied to visual tasks, the attention mechanism has attracted extensive attention from researchers in computer vision. Common attention mechanisms include the self-attention mechanism, channel attention mechanism, and spatial attention mechanism. The core of the attention mechanism is to highlight the most relevant and important features and to avoid being affected by irrelevant information, which is achieved by computing an attention map.

For example, DELF [37] designed a 2-layer CNN with an output layer of Softplus to calculate the attention scores of each position of the feature map on the penultimate layer of a fully-convolutional network to pick out the local features with the highest attention scores and to ignore the influence of some irrelevant backgrounds, which was more conducive to similarity matching between images. The attention module in DELG [39] adopted the strategy used in DELF [37]. Moreover, the attention mechanism in HoW [21] used a fixed attention mechanism not based on parameters, while the DELF attention module's parameters can be learned by end-to-end training. Whether the importance weights in the attention map are learnable serves as the primary criterion for determining whether it is parameter-based or not. To combine the Second-Order spatial information into the feature pool, Ng et al. directly integrated the Second-Order Attention module into the last two blocks of ResNet [42]. DOLG [38] reweighted the local features employing the self-attention technique in their model's local branch module to conduct orthogonal fusion with a single global feature. Weinzaepfel et al. have developed an iterative form of conventional attention by mapping the input features of variable data into an ordered set of N output features whose goal is to get rid of plenty of redundant local features [22]. In a nutshell, attention mechanisms provide the ability to draw attention to the most crucial and conspicuous parts of an image, which are widely used in visual tasks.

3 | MATERIALS AND METHODS

In this section, we will first introduce ℓ_2 -normalization and the use of cosine similarity to measure the similarity of image pairs and the state-of-the-art Generalized-Mean (GeM) pooling [20]. Then we present our model framework GS based on ResNet [42] and the method of descriptor whitening and query expansion. Finally, we introduce GSA, a model framework optimized by two-different loss functions.

3.1 | ℓ_2 -normalization and cosine similarity measure

3.1.1 | ℓ_2 -normalization:

Given a vector $V = \{v_1, v_2, \dots, v_d\}$ with dimension d used to represent an image descriptor, the vector after ℓ_2 -normalization is denoted as V norm, where $\|V\|_2^2$ represents the ℓ_2 -norm of vector V , then the following expression is defined as:

$$V_{norm} = \frac{V}{\|V\|_2}, \quad (1)$$

where $\|V\|_2^2 = \sqrt{v_1^2 + v_2^2 + \dots + v_d^2}$, and it is not difficult to figure out that the value of $\|V_{norm}\|_2^2$ is always 1.

3.1.2 | Cosine similarity measure

In general, the Euclidean distance can be used to measure the similarity of two images, but this method would involve the calculation of squares and surd form, which would be computationally complex. The solution to this problem is to replace the Euclidean distance with the cosine similarity since there is a negatively correlated monotonic relationship between them. Here, the ℓ_2 -normalization operation is applied if not otherwise stated. Given two vectors $V_1, V_2 \in \mathbb{R}^d$, the following expression can be obtained:

$$V_1^T V_2 \in [-1, 1]. \quad (2)$$

Then the cosine similarity $\cos\text{Sim}(\cdot, \cdot)$ is defined as follows:

$$\cos\text{Sim}(V_1, V_2) = \frac{V_1 * V_2}{\|V_1\|_2 \|V_2\|_2} = \sum_{i=1}^d V_{1i} * V_{2i}. \quad (3)$$

Therefore, at evaluation time, they can be interchanged to provide the same order of results:

$$\|V_1 - V_2\|_2^2 = \sqrt{2 - 2\cos\text{Sim}(V_1, V_2)}. \quad (4)$$

3.2 | Generalized-mean pooling

An input image $I \in \mathbb{R}^{b \times w \times 3}$ is given and fed to a fully-convolutional network (FCN) with its parameters denoted by θ . A feature map $f = \theta(I) \in \mathbb{R}^{H \times W \times C}$ is obtained as the output where H , W , and C represent the height, width, and channel of the feature maps after the last layer of convolution. Generalized-Mean pooling generates a final descriptor $D \in \mathbb{R}^{C \times 1}$ flexibly. Then D can be formalized as:

$$D = \left(\frac{1}{H * W} \sum_{(i,j) \in (H,W)} f_{(i,j,c)}^p \right)^{\frac{1}{p}}, \quad (5)$$

where $H > 0$, $W > 0$, and p is a trainable hyper-parameter. It can be inferred from Equation (5) that global max pooling happens when $p \rightarrow \infty$ and global average pooling occurs when $p = 1$. Thus, Equation (5) can be regarded as a trade-off between global average pooling and max pooling. The larger p is, the more localized the feature map responses will be. In the following trials, we follow [24, 39] and empirically set p to 3.0.

3.3 | GS model

In Figure 2, our proposed model GS—a ResNet [42] network-based model—is displayed. ResNet FCN, Generalized-Mean pooling (GeM), ℓ_2 -normalization, fully connected layer FC, and descriptor whitening are the major components of GS.

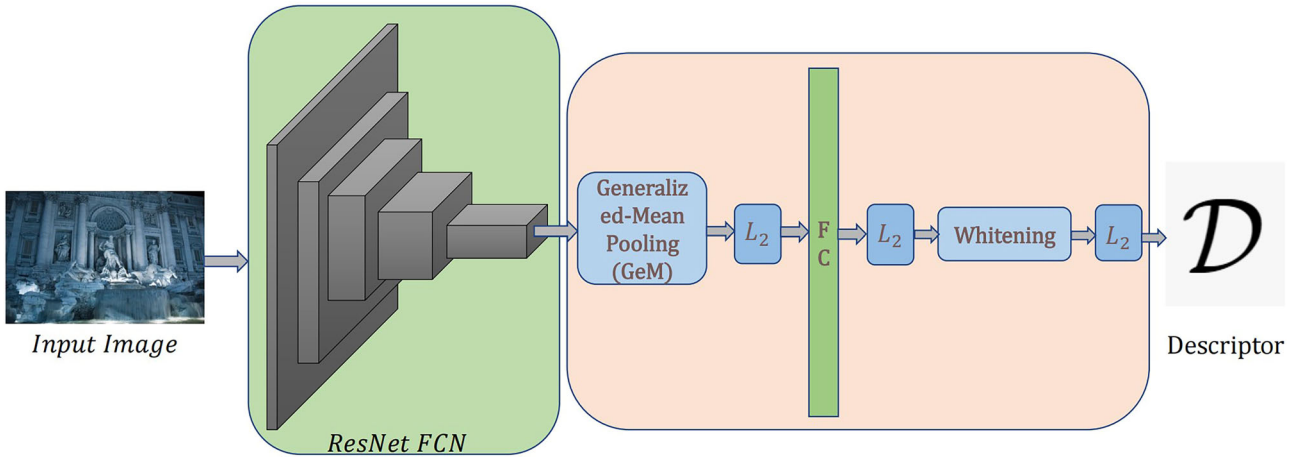


FIGURE 2 Block diagram of our proposed model (GS). An input image first goes through the fully-convolutional network of ResNet (removing the global average pooling and the fully-connected layers, etc.) to obtain a feature map, and then passes through GeM pooling, ℓ_2 -normalization, fully-connected (FC) layer, ℓ_2 -normalization, descriptor whitening process, and ℓ_2 -normalization, respectively. Finally, a descriptor of dimension 2048 is obtained.

The GS backbone network uses ResNet50 as an illustration. And after an input image passes through GS, a descriptor with dimension 2048 will be obtained. Note that the shape of the FC parameter is 2048×2048 .

During the training stage, Smooth-AP, a plug-and-play loss function by relaxing the indicator function of the non-differentiable AP with a sigmoid function, is adopted in our method. In this section, the Smooth-AP is reviewed as the necessary background.

3.3.1 | Objective definition

Given a query image I_q , the purpose is to pick out all images related to I_q in a retrieval dataset $D = \{I_1, I_2, \dots, I_N\}$ of length N and sort them in descending order of similarity. $Y(I_i, I_q)$ is denoted as 1 if I_i and I_q are similar, other-wise 0. The retrieval dataset is split into P_q and N_q for each query image I_q , indicating the sets of images that are related and unrelated to the query instance I_q , respectively. For instance, $A_1, A_2, \dots, Z_p$ in Figure 3 represents the retrieval dataset D of length P , for image A_1 , $P_{A_1} = \{A_1, A_2, \dots, A_p\}$ and $N_{A_1} = D - P_{A_1}$. It is worth noting that all samples in positive sample pools are selected randomly, which is a significant distinction from the hard sample mining strategy adopted for contrastive loss or triplet loss.

3.3.2 | Average precision (AP)

The value of the average precision is defined as the area under the P - R curve. Equation (6) can be employed to calculate the level of similarity between the query image I_q and all images in the retrieval dataset D .

$$S_D = \{s_i = \cosSim(V_q, V_i), i = 1, 2, \dots, N\}, \quad (6)$$

where $S_D = S_p \cup S_N$, V_q and V_i represent the corresponding descriptors of the query image I_q and the image I_i in the retrieval dataset. The following formula can be used to obtain the AP of the search image I_q :

$$AP_q = \frac{1}{|P_q|} \sum_{i \in S_p} \frac{\mathcal{R}(i, S_p)}{\mathcal{R}(i, S_D)}, \quad (7)$$

where $\mathcal{R}(i, S_p)$ and $\mathcal{R}(i, S_D)$ refer to the rankings of the instance I_i in the positive sample pool P and the retrieval dataset D , respectively. The ranking function $\mathcal{R}$ is defined as follows:

$$\mathcal{R}(i, S_D) = 1 + \sum_{j \in S_D, j \neq i} \mathbb{I}[s_j - s_i > 0], \quad (8)$$

where $\mathbb{I}[\cdot]$ denotes the indicator function, and if replacing D with P would indicate the similarity sequence between the query image I_q and its positive sample pool P . Thus, the AP of query instance I_q from Equation (7) can be re-expressed as follows:

$$AP_q = \frac{1}{|P_q|} \sum_{i \in S_p} \frac{1 + \sum_{j \in S_p, j \neq i} \mathbb{I}[s_j - s_i > 0]}{1 + \sum_{j \in S_D, j \neq i} \mathbb{I}[s_j - s_i > 0]}. \quad (9)$$

Since the indicator function $\mathbb{I}[\cdot]$ is not differentiable, which can not be back-propagated by gradient descent-based methods. A simple approach is to replace the indicator function $\mathbb{I}[\cdot]$ with a sigmoid function $\mathcal{G}(\cdot, \tau)$ (see Equation (10)), where τ is the temperature at which sharpness is regulated.

$$\mathcal{G}(x, \tau) = \frac{1}{1 + e^{-\frac{x}{\tau}}}. \quad (10)$$

The approximate AP of the query image I_q can be obtained by substituting the $\mathcal{G}$ function into the indicator function in

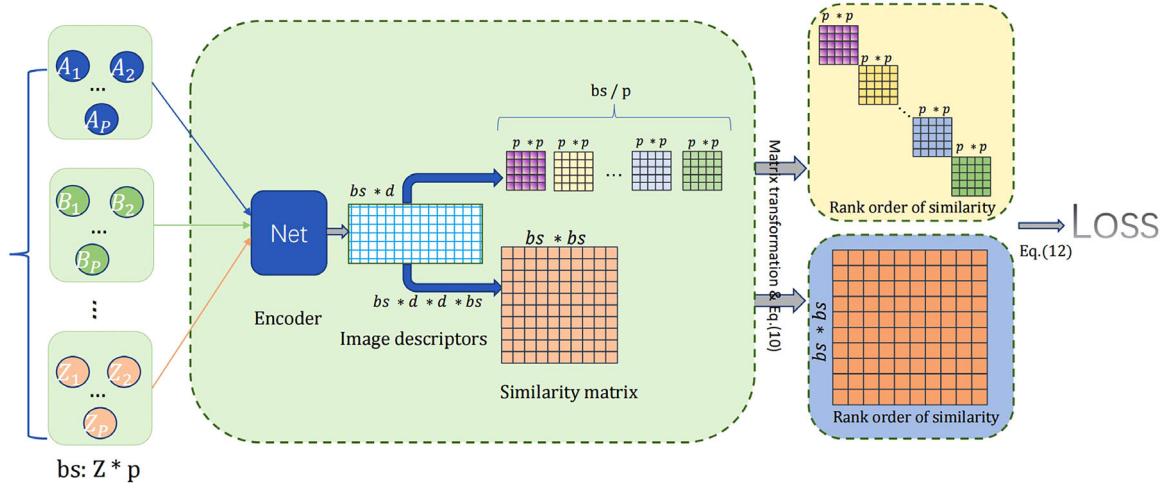


FIGURE 3 The overview of the training process. The bs denotes the mini-batch size, and the number of images (p) in each rectangular box within the opening brace is equal and of the same category. These images are fed into the network to produce a matrix of $bs \times d$. And then a simple matrix transformation is performed to obtain a Similarity matrix of $bs \times bs$ and bs/p Similarity matrices of $p \times p$. Finally, the operations of Equation (10) and a matrix operation are performed to obtain the same shape and number of matrices as in the previous step for calculating ℓ_{AP} .

Equation (9) as shown below:

$$AP_q \approx \frac{1}{|P_q|} \sum_{i \in S_p} \frac{1 + \sum_{j \in S_p} \mathcal{G}(s_j - s_i, \tau)}{1 + \sum_{j \in S_D} \mathcal{G}(s_j - s_i, \tau)}. \quad (11)$$

Naturally, the loss function during training can be formulated as follows:

$$\ell_{AP} = 1 - \frac{1}{N} \sum_{q=1}^N AP_q. \quad (12)$$

3.4 | Descriptor whitening and dimensionality reduction

Descriptor whitening has become an essential step in image retrieval since the work of Jecgou and Chum [43]. Their interpretation lies in jointly down grading co-occurrences and, thus, dealing with the problem of over-counting. The application of PCA-whitening to some methods [12, 14, 28] has gotten excellent results. In addition, Gordo et al. learned to whiten in an end-to-end manner, which was achieved by the vector offset and fully connected layers [23]. Our experiments follow [20] in which the CNN descriptor whitening is differentially learned from the same training dataset. This whitening technique will be applied as a post-processing step to whiten the descriptors once CNN has been fine-tuned. The two components of this whitening technique are whitening and rotation. The whitening part is the inverse of the square root of the covariance matrix within the class (matched image pairs) $C_S^{-\frac{1}{2}}$, where C_S is:

$$C_S = \sum_{Y(I_i, I_j)=1} (V_i - V_j)(V_i - V_j)^T. \quad (13)$$

The rotation part is the inter-class covariance matrix in the whitening space $\text{eig}(C_S^{-\frac{1}{2}} C_D C_S^{-\frac{1}{2}})$, where C_D is:

$$C_D = \sum_{Y(I_i, I_j)=0} (V_i - V_j)(V_i - V_j)^T. \quad (14)$$

Then the projection $P = C_S^{-\frac{1}{2}} \text{eig}(C_S^{-\frac{1}{2}} C_D C_S^{-\frac{1}{2}})$ is applied as $P^T(V_i - m)$ where m is the mean vector. To reduce the dimensionality of CNN descriptors to d , only the eigenvectors corresponding to the first d largest eigenvalues need to be taken. The projected vectors are subsequently ℓ_2 -normalized.

3.5 | Query expansion

An efficient query expansion technique called average query expansion (AQE) has been applied to some approaches [23, 28, 29] so as to further boost the accuracy of image retrieval. The new image descriptor V_q for a search instance I_q is obtained by averaging and ℓ_2 -normalizing the top n image descriptors from the returned retrieval results. The descriptor V_q can be described by the following formula:

$$V_q = \ell_2 \left(\sum_{i=1}^n \frac{1}{n} V_i \right). \quad (15)$$

The top n image descriptors by AQE have the same weight, which indicates that the ranking information of these images is not utilized as much as it should be. To this end, α -weighted query expansion (α is a hyper-parameter) has been proposed by Radenović et al. to address this problem [20]. Generally speaking, the higher the rankings of the returned retrieval

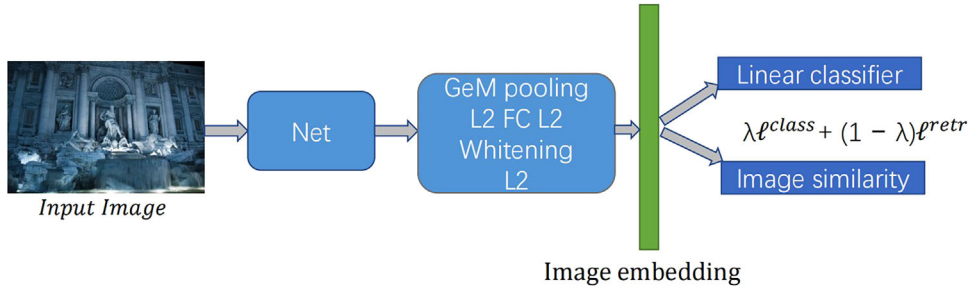


FIGURE 4 The overview of our proposed model GSA. Unlike the GS model, this model is trained jointly with two different loss functions. The Linear classifier is a fully connected layer of dimension $2048 \times n$ (n is the number of image classes), whose loss function is an Additive Angular Margin Loss named ArcFace used to calculate the classification loss. And Image similarity loss is the same as the loss function of GS.

results, the more similar they are to the query image I_q , and the greater the weight is given naturally. The specific formula is as follows:

$$V_q = \ell_2 \left(\sum_{i=1}^n (V_q^T V_i)^\alpha V_i \right). \quad (16)$$

3.6 | GSA model

Different loss functions for joint optimization have been successful [31, 32]. Inspired by them, we propose a model GSA trained with a joint loss function consisting of classification loss and instance retrieval loss (such as Smooth-AP), as shown in Figure 4. One of the loss functions is the same as the one in the GS model, namely ℓ_{AP} , and the other is an Additive Angular Margin Loss called ArcFace [44], which has achieved impressive results on large-scale human-face datasets. The joint loss function can be formulated as follows:

$$\ell_{GSA} = \lambda \ell_{AP} + (1 - \lambda) \ell_{AF} \quad (17)$$

where λ is a hyper-parameter, ℓ_{AP} is the equation from Equation (12), and ℓ_{AF} is defined as follows:

$$\ell_{AF} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j=1, j \neq y_i}^n e^{s \cos \theta_j}} \quad (18)$$

where N and n refer to the batch size and the number of categories of the images, respectively. We utilize $V_i \in \mathbb{R}^d$ to denote the deep feature of the i -th sample belonging to the y_i -th category. $W_j \in \mathbb{R}^d$ is depicted as the j -th column of the weight $W \in \mathbb{R}^{d \times n}$. For simplicity, we fix the bias $b_j = 0$ and convert the logit as $W_j^T V_i = \|W_j\| \|V_i\| \cos \theta_j$. Both the weight W and the deep features V are ℓ_2 -normalized, so $W_j^T V_i = \cos \theta_j$. In Equation (18), m is an additive angular margin penalty, s is a scaling factor, and $\theta_{y_i} = \arccos(\cos \theta_{y_i})$.

TABLE 1 Datasets used for training and evaluation

	Dataset	# Image	# Cluster
train	SfM-120k-train	91,642	551
validation	SfM-120k-val	6,403	162
test	ROxford(+R1M)	4,993(+R1M)/70	—
	RParis (+R1M)	6,332(+R1M)/70	—

4 | EXPERIMENTS

4.1 | Dataset

4.1.1 | Retrieval-SfM-120k

The SfM-120k training dataset is created by exploiting the geometry and camera positions from 3D models automatically reconstructed using the Structure-from-Motion (SfM) pipeline [20]. We utilize the SfM-120k dataset and follow the splitting scheme of the 551/162 3D model (cluster) for training and validation, respectively. Similar images are developed from visually overlapping (same 3D model) images, whereas the images between distinct 3D models are considered to be dissimilar.

4.1.2 | Revisited Oxford and Paris

Radenovic et al. have recently revised the Oxford and Paris building datasets by correcting labelling errors, adding new images as well as providing new evaluation protocols [45]. This evaluation protocol includes three levels of difficulty—Easy, Medium, and Hard. We only report the mAP for the Medium and Hard settings. Revisited Oxford (ROxford) and Revisited Paris (RParis) are now commonly used datasets for evaluating retrieval performance on large-scale image retrieval tasks. ROxford and RParis contain 4,993 and 6,332 images with 70 query images for each dataset, respectively. The supplementary 1M-distractor (R1M) dataset contains one million additional images used to test the robustness of our proposed models. The detailed information is shown in Table 1.

4.2 | Training setup and implementation details

All experiments are trained on the dataset retrieval-SfM-120k. We follow the authors' dataset partition strategy. In all experiments, ResNet50 [42] is applied as the backbone network initialized with ImageNet pre-trained weights. All the training images first undergo augmentations by randomly cropping and colour jittering; then they are resized to a maximum size of 362×362 , and their original aspect ratios are retained. But in the testing phase, we provide the network with authentic images without resizing and distortion. We use batch size of 224 to train our models on a single NVIDIA Tesla A100 GPU with 40G of DRAM. A total of 20 epochs of fine-tuning of the model are conducted for each epoch taking around one hour. The Adam optimizer is adopted by trying values 10^{-6} , 5×10^{-6} , 10^{-5} , 5×10^{-5} for the learning rate, with a learning rate exponential decay $l_0 * \gamma^{epoch}$, gamma 0.9, and a weight decay of 5×10^{-4} . For GeM pooling, the parameter p is fixed as 3.0. For the ArcFace loss, we empirically set the margin m to 0.15 and the scaling factor s to 30. We set α as 3 for the α -weighted query expansion (α QE) and α as 0 for the average query expansion (AQE). For λ in the joint loss function in the GSA model, which is set to 0.5. We set the τ in Equation (10) and the P in Figure 4 to 0.01 and 4, respectively. As for feature extraction, an image pyramid is used at the time of inference, multi-scale extraction is performed by re-scaling with coefficients 0.25, 0.353, 0.5, 0.707, 1.0, 1.414, 2.0. To obtain the final descriptor, we average the multi-scale descriptors and finally perform ℓ_2 -normalization.

4.3 | Evaluation metrics

We employ the mean Average Precision to measure the performance of our models. And the average precision of the query image I_q is represented as AP_{I_q} , as depicted in Equation (19):

$$AP_{I_q} = \frac{\sum_{i=1}^N P(i) * \mathbb{I}[i]}{R_{I_q}}, \quad (19)$$

where R_{I_q} denotes the number of images in the entire retrieved dataset of length N relevant to the query image I_q . $P(i)$ refers to the precision of the top i retrieved images, and $\mathbb{I}[\cdot]$ is an indicator function equal to 1 if the item in the retrieved result is relevant to the query image I_q and 0 otherwise. Then mean Average Precision is the mean value of the sum of AP of all query images and can be expressed as:

$$mAP = \frac{1}{Q} \sum_{q=1}^Q AP_{I_q}. \quad (20)$$

4.4 | Experimental results and analyses

To evaluate the performance of our proposed models GS and GSA, we assess them on the ROxf and RPar with an additional

1 M distractor images (400GB+). The metric for evaluation is mAP, and we now select the most classical models in image retrieval for performance comparison. The experimental results are depicted in Table 2.

To make each approach easier to grasp, we illustrate each method using abbreviations; CL, TL, and AP denote contrastive loss, triplet loss, and listwise loss, respectively. If no loss function is employed, then off-the-shelf(O) features are utilized. The results in Table 2 reveal that the retrieval performance obtained by employing pre-trained models for feature extraction of images is poor since they only take the global maximum (MAC(O)) or simply sum-up (SPoC(O)) from the last layer of feature maps as image descriptors. For instance, methods that employ off-the-shelf features have mAPs below 50% on ROxf-Medium, such as MAC(O), and SPoC(O). After R-MAC adopts the triplet loss, the retrieval performance improves in mAP by 11.1% on ROxf-Medium and 13.9% on ROxf-Hard; and by 4.9% on RPar-Medium and 7.3% on RPar-Hard. After optimizing triplet loss, the retrieval accuracy of the model is significantly boosted. Moreover, compared with the off-the-shelf GeM (GeM(O)), the GeM with contrastive loss (GeM(CL)) outperforms the GeM(O) with a boost of up to 19.7% on ROxf-Medium, 20.8% on ROxf-Hard, 6.5% on RPar-Medium and 7.6% on RPar-Hard. The above results demonstrate that the gains in ultimate retrieval performance by employing the triplet loss (TL) and the contrastive loss (CL) are significant. However, both the triplet loss and the contrastive loss must adopt the complex hard negative mining strategy (HMS), which needs to extract the features of the images in the training set to construct the hard sample pairs once an epoch of training is finished, so its disadvantage is quite time-consuming. GeM(AP) adopts the histogram box approximation method to approximate the non-differentiable AP function for end-to-end training. However, the loss function of GeM(AP) is complicated and it requires a large batch size (4096) and a fixed image size of 800×800 in the training set, which requires a large amount of video memory. Therefore, the GeM(AP) training process is divided into three stages to address the problem of insufficient video memory.

In contrast, the loss function used in our proposed model GS is simple to implement and only requires a batch size of 224 to achieve competitive performance. Compared to the GeM(AP), our GS outperforms the GeM(AP) with a boost of up to 1.8% on ROxf-Medium, 1.5% on ROxf-Hard, 2.2% on RPar-Medium, and 2.5% on RPar-Hard. In comparison to the GA, our GSA utilizes a joint loss function and outperforms the GS by up to 1.5% on RPar-Medium and 2.1% on ROxf-Hard.

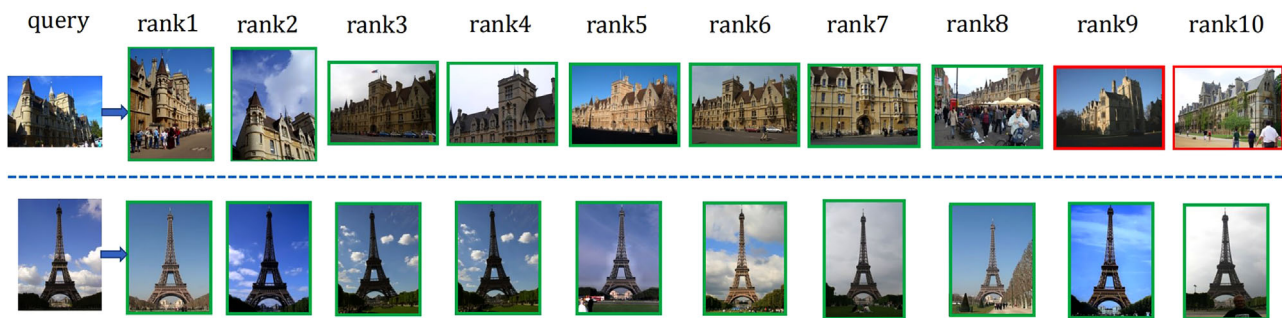
We also report retrieval results for the query expansion approach. Our models employ α QE with $n = 5$ on ROxf and AQE with $n = 50$ on RPar. Our GS model with QE outperforms other methods in six protocols out of the eight. We display the partial retrieval results in Figure 5, where the query images above and below the blue dashed line are from the thirteenth and sixth of the 70 query images in ROxf and RPar, respectively, as shown in detail in Figure 5.

In addition, to intuitively view the retrieval results, we have shown the information related to the retrieval results of

TABLE 2 Performance evaluation (mAP) for ROxford(+R1M) and RParis(+R1M).

Method	Medium				Hard			
	ROxf	ROxf+R1M	RPar	RPar+R1M	ROxf	ROxf+R1M	RPar	RPar+R1M
Local descriptors + ASMK matching								
HesAff-rSIFT-ASMK* [45]	60.4	45.0	61.2	42.0	36.4	25.7	34.5	16.5
HesAff-rSUFT-ASMK*+SP [36]	60.6	46.8	61.4	42.3	36.7	29.6	35.0	16.8
HesAff-HardNet-ASMK*+SP [46]	65.6	—	65.2	—	41.1	—	38.5	—
DELF-ASMK*+SP [37]	67.8	53.8	76.9	57.3	43.1	31.2	55.4	26.4
HOW [21]	78.3	63.6	80.1	58.4	55.8	36.8	60.1	30.7
Global descriptors + reranking with local features								
DLEG [39]	75.1	61.1	82.3	60.5	54.2	36.8	64.9	34.8
Global representations								
MAC(O) [13]	41.7	24.2	66.2	40.8	18.0	5.7	44.1	18.2
SPoC(O) [28]	39.8	21.5	69.2	41.6	12.4	2.8	44.7	15.3
CroW(O) [29]	42.4	21.2	70.4	42.7	13.3	3.3	47.2	16.3
R-MAC(O) [12]	49.8	29.2	74.0	49.3	18.5	4.5	52.1	21.3
R-MAC(TL) [23]	60.9	39.3	78.9	54.8	32.4	12.5	59.4	28.0
GeM(O) [20]	45.0	25.6	70.7	46.2	17.7	4.7	48.7	20.3
GeM(CL) [20]	64.7	45.2	77.2	52.3	38.5	19.9	56.3	24.7
GeM(AP) † [25]	65.5	—	77.6	—	41.0	—	57.1	—
GS(ours)	67.3	58.7	79.8	64.7	42.5	33.2	59.6	38.6
GSA(ours)	68.8	57.7	77.7	57.9	44.6	32.5	56.2	30.8
Query expansion								
R-MAC(TL) + α QE [23]	64.8	45.7	82.7	61.0	36.8	19.5	65.7	35.0
GeM(CL) + α QE [20]	67.2	49.0	80.7	58.0	40.8	24.2	61.8	31.0
GeM(AP) + α QE [25]	71.4	53.1	84.0	60.3	45.9	26.2	67.3	32.3
GS + QE ‡ (ours)	70.5	63.6	86.8	73.5	44.6	36.5	70.5	50.5
GSA + QE ‡ (ours)	73.4	62.2	84.6	66.3	49.7	35.9	67.9	42.0

Note: The global representations in our proposed models are learned from the ResNet50 backbone. The data in † are trained with ResNet50, the adopted QE in ‡ refers to α QE with n of 5 on ROxf and AQE with n of 50 on RPar. Abbreviations: (O) off-the-shelf features; (CL) fine-tuned with contrastive loss; (TL) triplet loss for fine-tuning; (AP) listwise average precision for fine-tuning; (SP) spatial validation.

**FIGURE 5** Demonstration of top-10 retrieved results. The retrieved results from the retrieval set of 1,005,994 images (ROxf+R1M) and the retrieval set of 1007323 images (RPar+R1M) are shown from top to bottom, respectively. Green and red boxes denote positive and negative images, respectively.

70 query images in Tables 3 and 4 for ROxford and RParis, respectively. We also demonstrate partial retrieval results in comparison with the classical GeM(CL), and the detailed results are shown in Figures 6 and 7.

4.5 | Ablation study

We conduct extensive ablation experiments and investigate the effect of various hyperparameter settings on the ultimate

TABLE 3 A comparison of the number of positive samples in the top min(10, All) images retrieved by our GS and GeM(CL) in ROxford and R1M

Index	Top	All	Index	Top	All	Index	Top	All	Index	Top	All	Index	Top	All	Index	Top	All
0	10/10	103	12	9/5	16	24	10/8	74	36	2/2	2	48	2/2	2	60	5/0	20
1	10/10	103	13	9/4	16	25	4/5	13	37	2/2	2	49	2/2	2	61	4/1	127
2	10/10	103	14	9/5	16	26	3/2	13	38	2/1	2	50	10/10	347	62	10/5	120
3	10/10	103	15	10/1	24	27	4/2	13	39	2/2	2	51	10/10	347	63	0/0	123
4	10/10	103	16	10/8	24	28	6/5	13	40	10/10	104	52	10/10	347	64	10/0	33
5	10/4	20	17	7/6	24	29	4/5	13	41	10/10	104	53	10/10	347	65	10/0	33
6	10/6	20	18	9/0	24	30	10/10	41	42	10/10	104	54	10/10	347	66	0/0	31
7	6/7	20	19	9/0	24	31	10/10	41	43	10/9	104	55	10/10	103	67	2/5	332
8	10/8	20	20	10/10	85	32	10/10	41	44	10/10	104	56	10/0	97	68	10/4	308
9	10/6	20	21	10/10	85	33	10/10	41	45	2/2	2	57	5/3	110	69	10/5	294
10	8/7	16	22	10/10	85	34	10/10	41	46	2/2	2	58	10/2	20			
11	1/1	2	23	10/8	85	35	2/2	2	47	2/2	2	59	8/0	15			

Note. (Index) the index of query images; (All) the number of positive samples of the query image, which may be less than 10; (Top) the a and b in a/b denote the number of positive samples in the top min (10, All) of the retrieval results of our model GS and GeM(CL), respectively.

TABLE 4 A comparison of the number of positive samples in the top 10 images retrieved by our GS and GeM(CL) in RParis and R1M

Index	Top	All	Index	Top	All	Index	Top	All	Index	Top	All	Index	Top	All	Index	Top	All
0	10/10	199	12	10/10	311	24	10/10	328	36	10/10	178	48	10/10	232	60	10/10	619
1	10/10	199	13	10/10	311	25	10/10	83	37	10/10	178	49	10/10	232	61	10/10	156
2	10/10	199	14	10/10	311	26	10/7	83	38	10/10	178	50	7/9	250	62	6/4	125
3	10/10	199	15	10/10	216	27	10/10	83	39	10/10	178	51	9/9	251	63	10/6	125
4	10/10	199	16	10/10	216	28	10/10	83	40	10/10	76	52	9/10	250	64	7/4	173
5	10/10	636	17	10/10	216	29	10/10	83	41	10/10	76	53	9/10	251	65	10/4	173
6	10/10	636	18	10/10	216	30	10/10	172	42	10/10	76	54	10/10	250	66	10/9	157
7	10/10	636	19	10/10	216	31	10/10	172	43	10/10	76	55	10/9	190	67	8/8	215
8	10/10	636	20	10/10	328	32	10/10	172	44	10/10	76	56	9/7	187	68	7/4	215
9	10/10	636	21	10/10	328	33	10/10	172	45	10/10	232	57	9/8	164	69	8/5	193
10	10/10	311	22	10/10	328	34	10/10	172	46	10/10	232	58	8/9	613			
11	10/10	311	23	10/10	328	35	10/10	178	47	10/10	232	59	4/9	512			

Note. (Index) the index of query images; (All) the number of positive samples of the query image; (Top) the a and b in a/b denote the number of positive samples in the top 10 of the retrieval results of our model GS and GeM(CL), respectively.

retrieval accuracy, such as descriptor dimensionality and descriptor whitening. In addition to the mAP metric, mP@10 is also employed as a measure of retrieval accuracy in the ablation experiments. P@10 refers to the precision of the top ten images in the retrieval results, and mP@10 refers to the mean precision over the entire query images. Our standard experiment based on GS takes descriptors with a dimensionality of 2048, batch size of 224, maximum training image size of 362, retrieval-SfM-120k whitening, and no query expansion. In addition, we also explore the effect of the margin m and the scaling factor s on the final retrieval performance.

4.5.1 | Effect of descriptor dimensionality

In general, the higher the dimensionality of a descriptor is, the richer the information it contains. However, higher dimen-

sionality takes up more storage foot-print, a dimensionality reduction operation (see Section 3.4) can be performed (in the case of losing a small amount of accuracy) in exchange for less storage footprint. The detailed results are shown in Table 5.

4.5.2 | Effect of whitening

Descriptor whitening processing is considered an indispensable part of the image retrieval domain. The improvement of retrieval performance by whitening is significant, as shown in Table 6. The mAP has been improved with a boost up to 6.37% on ROxf-Medium, 6.69% on ROxf-Hard while the mP@10 has been improved with a boost up to 6.43% and 6.86% on ROxf-Medium and ROxf-Hard respectively by employing retrieval-SfM-120k whitening. We argue that the effects of whitening, in addition to the one described in [43], are to (1)

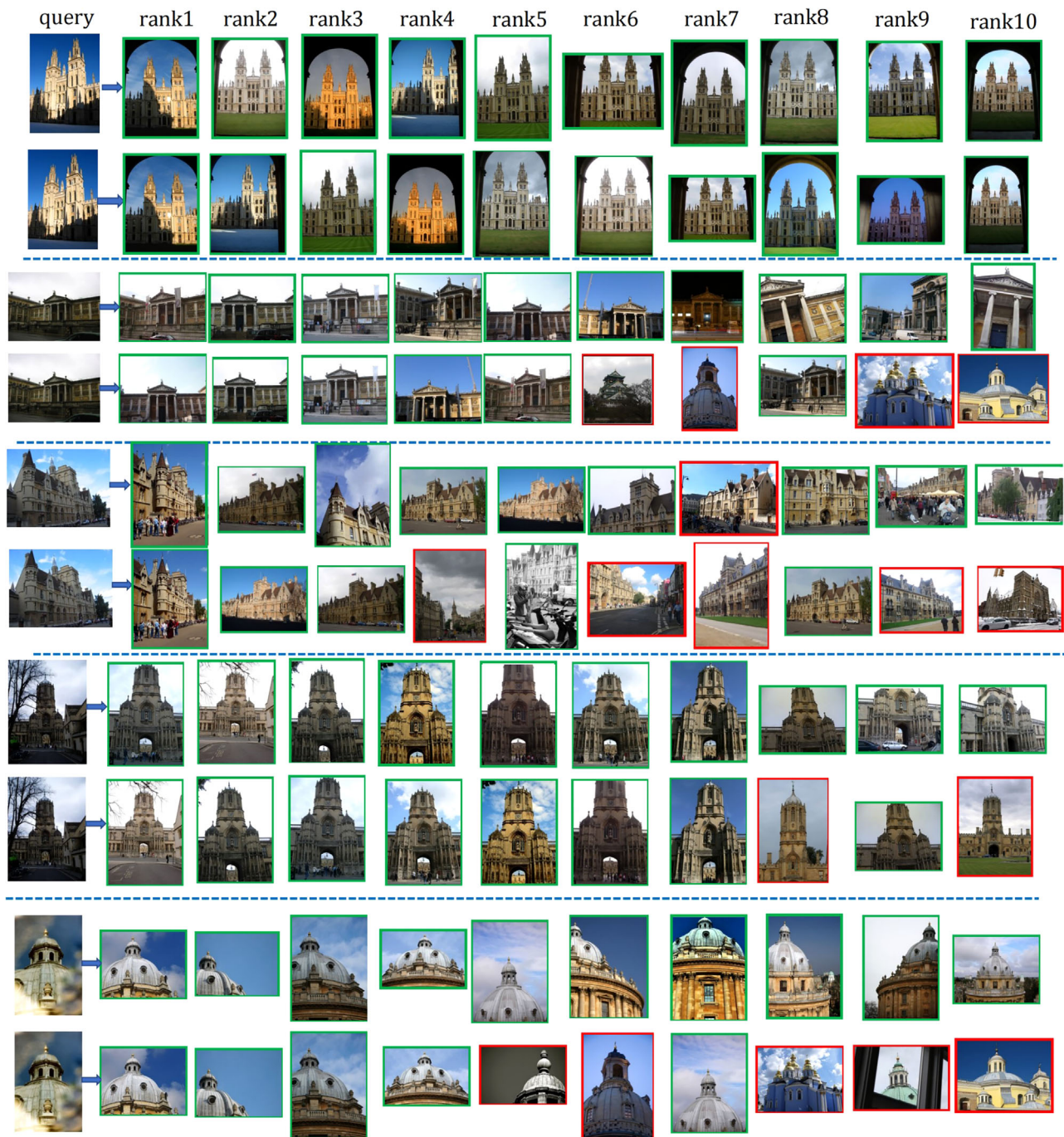


FIGURE 6 Demonstration of top-10 retrieved results. The retrieved results from the retrieval set of 1,005,994 images (ROxf+R1M). The top and bottom are the retrieval results of our model GS and GeM(CL), respectively. These five retrieved images are indexed as 0, 9, 14, 24, and 69, respectively. Green and red boxes denote positive and negative images, respectively.

de-correlate the various dimensions of the data and (2) make the variance of each dimension of the descriptors 1.

4.5.3 | Effect of batch size

The loss function of the directly optimizing AP is strongly influenced by the batch size, as discussed in Section 5. Theoretically, the larger the batch size is, the more discriminative and robust

the features are. Table 7 shows that large batch size results in better mAP.

4.5.4 | Effect of image size

The larger the image size is, the richer the texture information contains in the image. The maximum size of the raw image in the training set is 1024 pixels, and we downsample to a

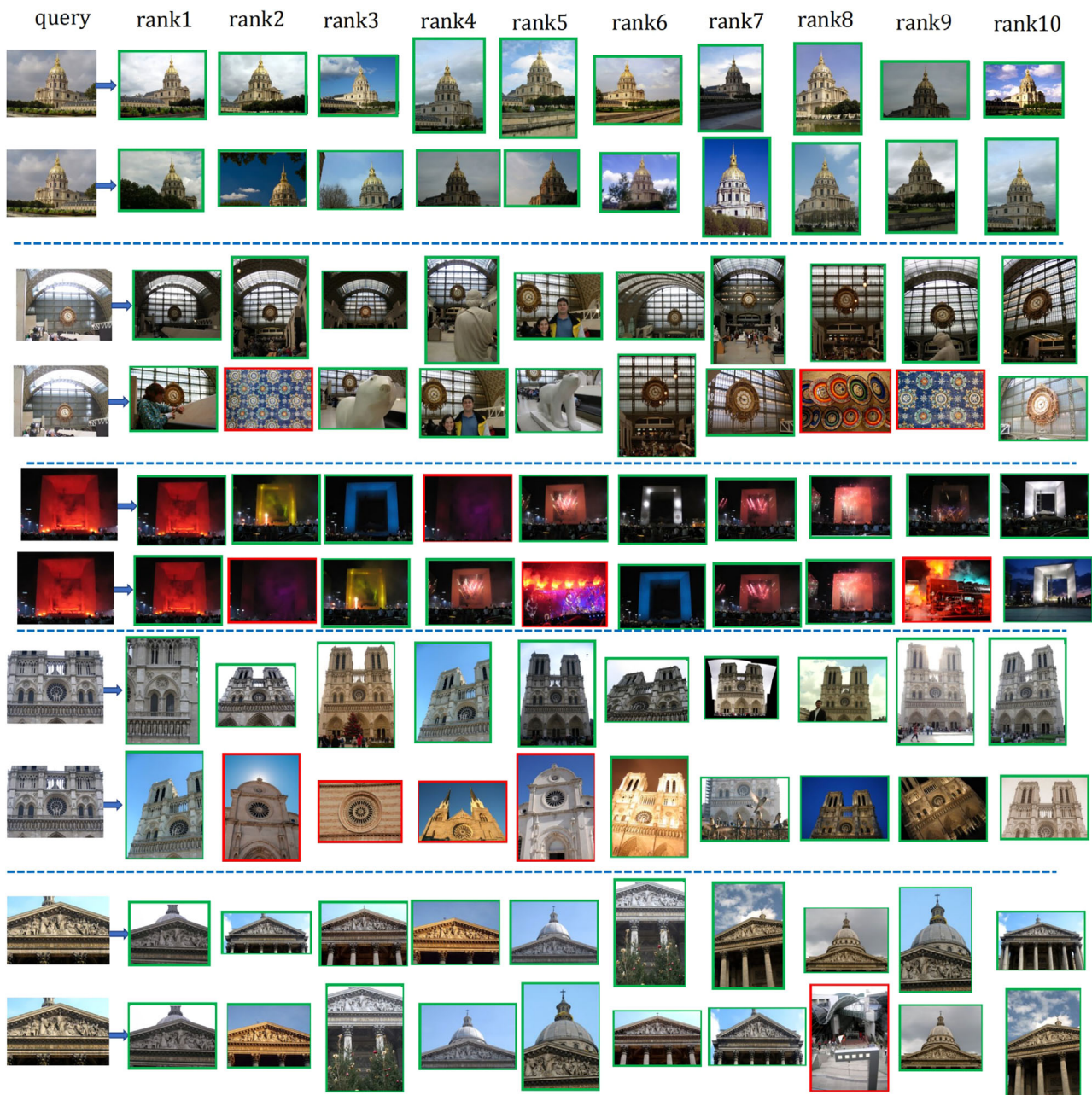


FIGURE 7 Demonstration of top-10 retrieved results. The retrieved results from the retrieval set of 1007323 images (RPar+R1M). The top and bottom are the retrieval results of our model GS and GeM(CL), respectively. These five retrieved images are indexed as 10, 26, 56, 63, and 66, respectively. Green and red boxes denote positive and negative images, respectively.

TABLE 5 The effect of various dimensions of image features on retrieval performance

Dimensionality	ROxford				RParis			
	Medium		Hard		Medium		Hard	
	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10
2048	67.34	87.29	42.46	53.86	79.81	98.43	59.58	89.29
1024	65.38	85.23	40.59	52.14	77.92	98.14	56.53	89.14
512	63.19	84.0	39.29	50.29	75.82	98.0	52.82	87.71

TABLE 6 The effect of whitening on retrieval performance

whitening	ROxford				RParis			
	Medium		Hard		Medium		Hard	
	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10
w/o whitening	60.97	80.86	35.77	47.0	73.16	96.86	48.81	83.86
Retrieval-SfM-30k	66.45	86.57	41.01	51.57	79.16	98.43	59.0	89.71
Retrieval-SfM-120k	67.34	87.29	42.46	53.86	79.81	98.43	59.58	89.29

TABLE 7 The effect of different batch sizes on retrieval performance

Batch size	ROxford				RParis			
	Medium		Hard		Medium		Hard	
	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10
112	66.90	86.68	41.63	53.07	79.60	98.49	59.10	89.69
224	67.34	87.29	42.46	53.86	79.81	98.43	59.58	89.29

TABLE 8 The effect of different image sizes on retrieval performance

Image size	ROxford				RParis			
	Medium		Hard		Medium		Hard	
	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10
224	65.71	86.29	40.47	52.0	78.30	98.86	59.92	89.29
362	67.34	87.29	42.46	53.86	79.81	98.43	59.58	89.29

maximum size of 362 or 224 pixels instead. Overall, we observe that a large value (362) results in higher mAP scores than a small value. The particular retrieval results are displayed in Table 8.

4.5.5 | Effect of query expansion

Query expansion has been already explained in Section 3.5. Query expansion is a simple yet efficient approach that can improve retrieval accuracy. As shown in Table 9, α QE and AQE have been applied to our methods. And we can observe that the method with α QE of $n = 5$ improves in mAP by 3.12% on ROxf-Medium and 2.18% on ROxf-Hard than the method without QE; while the approaches with AQE on ROxf degrades the retrieval performance. As for the RPar dataset, the method with AQE of $n = 50$ improves in mAP by 6.94% and 10.87% on RPar-Medium and RPar-Hard, respectively. Surprisingly, the retrieval performance is also boosted when applying α QE to it. We speculate that the discrepancies between these two datasets result in different increase rates in retrieval accuracy for ROxf and RPar achieved by the distinct query expansion techniques.

4.5.6 | Effect of the margin m and the scaling factor s

Specifically, we set up four groups of experiments with $s = 40.0$ $m = 0.15$, $s = 20.0$ $m = 0.15$, $s = 30.0$ $m = 0.10$ and $s = 30.0$ $m = 0.20$, respectively. The detailed results of the experiments can be seen in Table 10. We combine the experimental data in the GSA row ($s = 30$, $m = 0.15$) in Table 2 and the retrieval results in Table 10 to observe that the retrieval results in Table 10 are all lower than the data in that row of GSA, and we speculate that this may be the result of the margin loss's authors have verified through extensive experiments that $s = 30$ $m = 0.15$ is optimal.

5 | DISCUSSION AND LIMITATIONS

Here, we provide some discussion of the current implementation and the complexity of the model. The storage foot-print of the extracted R1M distractor images is around 8GB. The similarity score matrix can be obtained by multiplying the feature matrix of the query images with the feature matrix of the retrieval dataset during the evaluation, which allows the

TABLE 9 The effect of distinct approaches of query expansion on retrieval performance

QE	ROxford				QE	RParis			
	Medium		Hard			Medium		Hard	
	mAP	mP@10	mAP	mP@10		mAP	mP@10	mAP	mP@10
w/o QE	67.34	87.29	42.46	53.86	w/o QE	79.81	98.43	59.58	89.29
AQE/ $n=5$	63.71	81.52	39.62	58.48	AQE/ $n=5$	86.43	98.29	69.86	95.0
AQE/ $n=10$	62.69	79.38	38.76	52.56	AQE/ $n=10$	86.75	98.57	70.45	95.29
α QE/ $n=5$	70.46	88.67	44.64	58.57	α QE/ $n=5$	85.23	98.71	67.90	93.43
α QE/ $n=10$	68.33	85.43	43.47	57.63	α QE/ $n=10$	85.35	98.86	68.20	93.29

TABLE 10 The effect of the margin m and the scaling factor s on retrieval performance

s/m	ROxford				RParis			
	Medium		Hard		Medium		Hard	
	mAP	mP@10	mAP	mP@10	mAP	mP@10	mAP	mP@10
$s = 40, m = 0.15$	68.23	87.14	44.12	57.14	77.38	97.57	55.68	90.43
$s = 20, m = 0.15$	67.43	86.71	43.33	55.14	76.33	97.21	54.25	89.86
$s = 30, m = 0.10$	68.08	87.14	43.92	56.57	77.21	98.14	55.61	89.86
$s = 30, m = 0.20$	68.11	87.29	44.26	56.71	76.72	97.86	54.75	90.71

calculation of the mAP. In addition, the retrieval speed is relatively fast as we do not perform operations such as spatial validation. Instead of employing a loss function based on local loss metric learning (in the fashion of pairs or triples), our models are based on an AP loss function, that is, more samples are considered for training simultaneously. As a consequence, the trained model has better robustness and more stable retrieval performance. In short, our current model is sufficiently simple yet effective. Although some methods [21, 39] that combine global and local features achieve impressive retrieval performance, we argue that they exist the following problems: (1) Both contrastive loss and triplet loss require the use of HMS, which results in higher complexity of the training process; (2) the adoption of local features takes up a large amount of storage footprint, for instance, the storage of the feature vectors of ROxf(or RPar)+R1M in DELG [39] takes up 400GB+; (3) the use of local features for re-ranking degrades the speed of retrieval.

There are also limitations to our models. The listwise loss function of the direct optimization AP requires a large batch size so that the trained model is more capable of extracting discriminative features. However, the performance of the model only depends on the batch size that a single GPU can accommodate. In addition, implementing Smooth-AP only needs to consider penalizing negative instances before positive ones to remove redundant representations from the loss function.

Therefore, it is essential to design superior loss functions that directly optimize AP and tackle the problems caused by the batch size in the future. Besides, we believe that image retrieval can achieve competitive retrieval performance with only image-

level labels (or without image-level annotations, namely, in an unsupervised manner) and simple loss functions without the necessity of complex HMS.

6 | CONCLUSION

Here, we make the first attempt to combine the state-of-the-art GeM pooling with a direct optimization of the AP loss function and propose our GS model that requires only image-level labels for end-to-end training without relying on a complex HMS employing metric learning. Our model achieves efficient single-stage image retrieval with high speed and only a modest storage footprint (around 8GB) is required for the extracted image features. We also attempt to engineer the model GSA utilizing two loss functions for joint optimization, which achieves the best performance on the ROxf dataset. In addition, we have applied different approaches of query expansion to further improve the retrieval accuracy of the model at the cost of retrieval time. Extensive ablation experiments have also demonstrated the effect of each component on the final retrieval performance. Our GS model outperforms in mAP the GeM(AP) by 1.8% on ROxf-Medium, 1.5% on the ROxf-Hard; and by 2.2% on RPar-Medium, 2.5% on the RPar-Hard, and the experimental results demonstrate the effectiveness of our proposed model.

AUTHOR CONTRIBUTIONS

Jinliang Yao: Supervision, investigation and methodology. Yongqing Li: Methodology, resources, original draft and software. Bing Yang: Formal analysis, software and review. Chenrui Wang: Methodology, writing and original draft.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in train dataset at <http://cmp.felk.cvut.cz/cnnimageretrieval/data/>. The test datasets ROxford5k, RParis6k, and R1M are in <http://www.robots.ox.ac.uk/~vgg/data/oxbuildings>, <http://www.robots.ox.ac.uk/~vgg/data/parisbuildings>, and <http://ptak.felk.cvut.cz/revisitop/revisitop1m/jpg/>, respectively.

ORCID

Yongqing Li  <https://orcid.org/0000-0002-6254-907X>

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